

# Multiple Linear Regression — Fitting

Jericho Lawson

MATH 240

Week 6, Lecture #1 — Ch. 8

# Types of Regression

- **Simple linear regression:** Modeling some explanatory variable  $X$  to attempt to explain response variable  $Y$
- **Multiple linear regression:** Modeling multiple explanatory variables  $X_1, X_2, \dots, X_p$  to attempt to explain response variable  $Y$
- **Logistic regression:** Modeling multiple explanatory variables  $X_1, X_2, \dots, X_p$  to attempt to explain some binary response  $Y$

# Motivations for MLR

We might be curious about whether a relationship exists between multiple explanatory variables and the response. Examples include:

- 1 Does the amount of time spent in **daycare** before kindergarten and the **quality of teaching** affect the probability of showing **behavior problems** in elementary school?
- 2 Is **average income** and **police presence** associated with **crime rate** in different districts?
- 3 Is **coffee consumption** in an office and the **number of breaks** associated with the **growth of revenue**?

In multiple linear regression (MLR), we have more variables to examine that could potentially affect the outcome.

# Objectives

- **Use** R to fit a least-squares line in multiple linear regression
- **Conduct** model selection using backward elimination and forward selection
- **Diagnose** various plots to verify conditions for a least-squares fit
- **Complete** other goals of linear regression, such as interpretation and prediction

# Least-Squares Line for Multiple Linear Regression

We still minimize the sum of squared errors (i.e.  $\sum e_i^2$ ) for all data points in order to get the following least-squares line:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p$$

- $\hat{\beta}_0$ : estimated intercept term
  - ▶ Represents value of  $\hat{y}$  when  $x = 0$
- $\hat{\beta}_i$ : estimated slope for covariate  $x_i$ 
  - ▶ Represents the change in  $\hat{y}$  for every one unit increase in  $x_i$  (while keeping all other variables constant)

# Revised Steps for Multiple Linear Regression

We may want to do the following tasks when performing multiple linear regression:

- 1 Exploratory data analysis (EDA)
- 2 Linear regression fitting
- 3 **Model Selection**
- 4 Diagnostics
- 5 Interpretation and hypothesis testing of coefficients
- 6 Prediction

# Tutorial: duke\_forest Dataset - Introduction

- “Data on houses that were recently sold in the Duke Forest neighborhood of Durham, NC in November 2020” (OpenIntro)
- Number of observations  $n = 98$ ; number of features  $p = 13$

You can then use the following code to import the ‘openintro’ library and read in the duke\_forest data:

- `library(openintro)`
- `data(duke_forest)`
- `duke_forest = duke_forest[!is.na(duke_forest$lot), ]`
- `head(duke_forest)`

# Tutorial: duke\_forest Dataset - Introduction

```
> head(duke_forest)
```

```
# A tibble: 6 × 13
```

|   | address<br><chr>                      | price<br><dbl> | bed<br><dbl> | bath<br><dbl> | area<br><dbl> | type<br><chr> | year_built<br><dbl> | heating<br><chr> | cooling<br><fct> |
|---|---------------------------------------|----------------|--------------|---------------|---------------|---------------|---------------------|------------------|------------------|
| 1 | 1 Learned Pl, Durham, NC 27705        | 1.52e6         | 3            | 4             | 6040          | Sing...       | 1972                | Other,...        | central          |
| 2 | 1616 Pinecrest Rd, Durham, NC 27705   | 1.03e6         | 5            | 4             | 4475          | Sing...       | 1969                | Forced...        | central          |
| 3 | 2418 Wrightwood Ave, Durham, NC 27705 | 4.20e5         | 2            | 3             | 1745          | Sing...       | 1959                | Forced...        | central          |
| 4 | 2527 Sevier St, Durham, NC 27705      | 6.80e5         | 4            | 3             | 2091          | Sing...       | 1961                | Heat p...        | central          |
| 5 | 2218 Myers St, Durham, NC 27707       | 4.29e5         | 4            | 3             | 1772          | Sing...       | 2020                | Forced...        | central          |
| 6 | 2619 Vesson Ave, Durham, NC 27707     | 4.56e5         | 3            | 3             | 1950          | Sing...       | 2014                | Forced...        | central          |

Questions to ask:

- How does the number of bedrooms affect the price of a house?
- Does the age of the house affect the price of a house, and if so, to what extent?
- Does having central cooling shift the price of a house?

With the given variables, we will determine which variables explain the price of a house.

# Exploratory Data Analysis (EDA)

- Identifying **general patterns** in the data
- For MLR: we are concerned about **collinearity**
  - ▶ i.e. we do not want too much correlation between predictor variables (redundancy)
  - ▶ e.g. height and arm span of a person to predict throwing ability
- Can be in the form of:
  - ▶ Plots and graphs (e.g. scatter plots, histograms)
    - ★ Between predictor variables; between a response and predictor
  - ▶ Summary tables (e.g. five-number summaries)
- Remember to keep the rules for effective communication in mind

# Fitting

As seen in simple linear regression, we can use the `lm()` function to fit a multiple linear regression model:

- `lm(response ~ x1 + x2 + ... + xp)`

The code will find a least-squares line in the following format (that reduces the sum of squared residuals  $\sum \epsilon_i^2 = \sum (y_i - \hat{y}_i)^2$ ):

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p$$

# Interpretation

For interpretation of this least-squares line:

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \cdots + \hat{\beta}_p x_p,$$

- For  $\beta_0$ : value of response while letting all  $x_i = 0$
- For  $\beta_i$ : average change in response for every one unit increase in  $x_i$   
holding all other variables constant

# Tutorial: Duke Forest Dataset - Fitting

We can fit the model with the following command:

- `mod1 = lm(price ~ bed + bath + area + year_built + cooling + lot, data = duke_forest)`

# Tutorial: Duke Forest Dataset - Fitting

We can then use the following code to get a summary of our fit:

- `summary(mod1)`

Call:

```
lm(formula = price ~ bed + bath + area + year_built + cooling +  
    lot, data = duke_forest)
```

Residuals:

| Min     | 1Q     | Median | 3Q    | Max    |
|---------|--------|--------|-------|--------|
| -723283 | -78169 | -1185  | 85504 | 383447 |

Coefficients:

|                | Estimate   | Std. Error | t value | Pr(> t )     |
|----------------|------------|------------|---------|--------------|
| (Intercept)    | -2.911e+06 | 1.788e+06  | -1.628  | 0.10703      |
| bed            | -1.369e+04 | 2.593e+04  | -0.528  | 0.59874      |
| bath           | 4.108e+04  | 2.466e+04  | 1.666   | 0.09928      |
| area           | 1.021e+02  | 2.313e+01  | 4.416   | 2.79e-05 *** |
| year_built     | 1.459e+03  | 9.140e+02  | 1.597   | 0.11385      |
| coolingcentral | 8.407e+04  | 3.034e+04  | 2.771   | 0.00679 **   |
| lot            | 3.561e+05  | 7.594e+04  | 4.690   | 9.70e-06 *** |

---

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 145100 on 90 degrees of freedom

Multiple R-squared: 0.6153, Adjusted R-squared: 0.5896

F-statistic: 23.99 on 6 and 90 DF, p-value: < 2.2e-16

# Tutorial: Duke Forest Dataset - Interpretation

Using the information from the `summary()` command, we can write the least-squares line as:

$$\begin{aligned}\hat{y} = & -2,910,715 - 13,692x_{bed} + 41,076x_{bath} + \\ & 102.13x_{area} + 1,459.2x_{year\_built} + \\ & 84,065x_{cooling=central} + 356,140x_{lot}\end{aligned}$$

**Interpretation for  $x_{bath}$ :** Holding all other variables constant, for every additional bathroom added inside a Duke home, there is an average increase of \$41,076 in a home's sale price.

# Tutorial: Duke Forest Dataset - Interpretation

We can calculate the residuals for all observations in the dataset:

- `mod1$residuals`

Suppose we wanted to manually calculate the residual of the fifth observation, which has the following information:

| price         | bed   | bath  | area        | year_built  | cooling | lot   |
|---------------|-------|-------|-------------|-------------|---------|-------|
| <dbl>         | <dbl> | <dbl> | <dbl>       | <dbl>       | <fct>   | <dbl> |
| <u>428500</u> | 4     | 3     | <u>1772</u> | <u>2020</u> | central | 0.16  |

# Tutorial: Duke Forest Dataset - Interpretation

With the following values for the covariates above, we can substitute them in the model below:

$$\begin{aligned}\hat{y}_5 &= -2,910,715 - 13,692(4) + 41,076(3) + \\ &\quad 102.13(1772) + 1,459.2(2020) + \\ &\quad 84,065(1) + 356,140(0.16) \\ &= 427,350.8 \quad \text{using rounded coefficients} \\ &= 427,451.0 \quad \text{actual fitted value}\end{aligned}$$

From here, we can calculate the residual as such:

$$\begin{aligned}e_5 &= y_5 - \hat{y}_5 \\ &= 428500 - 427451 \\ &= 1049\end{aligned}$$

## Activity: Duke Forest Dataset - Interpretation

**Activity:** Using the 11th observation, please find the residual using the command `mod1$residuals[11]` and by-hand with the model generated in R.

# Tutorial: Duke Forest Dataset - Goodness-of-Fit

The goodness-of-fit can be calculated with the adjusted  $R^2$  value, as seen below:

Multiple R-squared: 0.6153,      Adjusted R-squared: 0.5896

Recall that the multiple and adjusted  $R^2$  values can be calculated as such (with  $n$  = number of obs. and  $k$  = number of variables):

$$\text{Multiple } R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

$$\text{Adjusted } R^2 = 1 - \frac{\left(\frac{SSE}{n-k-1}\right)}{\left(\frac{SST}{n-1}\right)}$$

- SSE: sum of squared errors/residuals;  $\sum_{i=1}^n e_i^2 = \sum_{i=1}^n (y_i - \hat{y}_i)^2$
- SSR: sum of squares for regression;  $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$
- SST: sum of squares total;  $\sum_{i=1}^n (y_i - \bar{y})^2$

# Tutorial: Duke Forest Dataset - Goodness-of-Fit

- Recall that  $R^2$  helps us determine the amount of variability explained by the model
- $R^2$  will always increase when we include more variables in the model, even if the variables are not useful
- The adjusted  $R^2$  value is a measure that helps us balance model efficacy with model size

# Problem: Goodness-of-Fit

We can calculate the SSE and SST with the following code:

- `sum(mod1$residuals2)`
- `sum((duke_forest$price - mean(duke_forest$price))2)`

With the code above, we get the following:

- $SSE = 1.895747 \times 10^{12}$
- $SST = 4.930202 \times 10^{12}$

Using the formulas mentioned in the last slide, **calculate the multiple and adjusted  $R^2$  values.**

# For Wednesday!

- Read chapter 8 of IMS
- Homework #4 has been assigned
  - ▶ Involves textbook problems and an R question
  - ▶ Due on Sunday at 11:59pm
- On your own time: Fit a model with 3 predictors from the `mtcars` dataset using `mpg` as your response.
  - ▶ Write the least-squares line.
  - ▶ Interpret each of the  $\beta$  coefficients.
- Checkpoint #3 during Friday's class
  - ▶ 15 minute warm-up quiz at beginning of class